

Class10

Kate Ruiz (PID: A17671200)

2

```
pdb_file <- ("pdb_stats.csv")
pdb_f = read.csv(pdb_file, row.names=1)
head(pdb_f)
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	178795	21825	12773	343	226	84
Protein/Oligosaccharide	10363	3564	34	8	11	1
Protein/NA	9106	6335	287	24	7	0
Nucleic acid (only)	3132	221	1566	3	15	3
Other	175	25	33	4	0	0
Oligosaccharide (only)	11	0	6	0	1	0
	Other	Total				
Protein (only)	32	214078				
Protein/Oligosaccharide	0	13981				
Protein/NA	0	15759				
Nucleic acid (only)	1	4941				
Other	0	237				
Oligosaccharide (only)	4	22				

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
sum_xray <- sum(pdb_f$X.ray)
sum_xray
```

[1] 201582

```
sum_em <- sum(pdb_f$EM)
sum_em
```

```
[1] 31970
```

```
sum_total <- sum(pdb_f$Total)
sum_total
```

```
[1] 249018
```

```
percent_xray_em <- ((sum_xray + sum_em)/(sum_total)) * 100
percent_xray_em
```

```
[1] 93.7892
```

```
percent_xray <- ((sum_xray)/(sum_total)) * 100
percent_xray
```

```
[1] 80.95077
```

```
percent_em <- ((sum_em)/(sum_total)) * 100
percent_em
```

```
[1] 12.83843
```

The answer is that in combination they make up 93.79%. Individually, x-rays make up 80.95% and EMs make up 12.84%.

Q2: What proportion of structures in the PDB are protein?

```
(178795 + 10363 + 9106)/sum_total
```

```
[1] 0.7961834
```

The proportion is 0.80.

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

I believe I saw 20.

3 Visualizing the HIV-1 protease structure

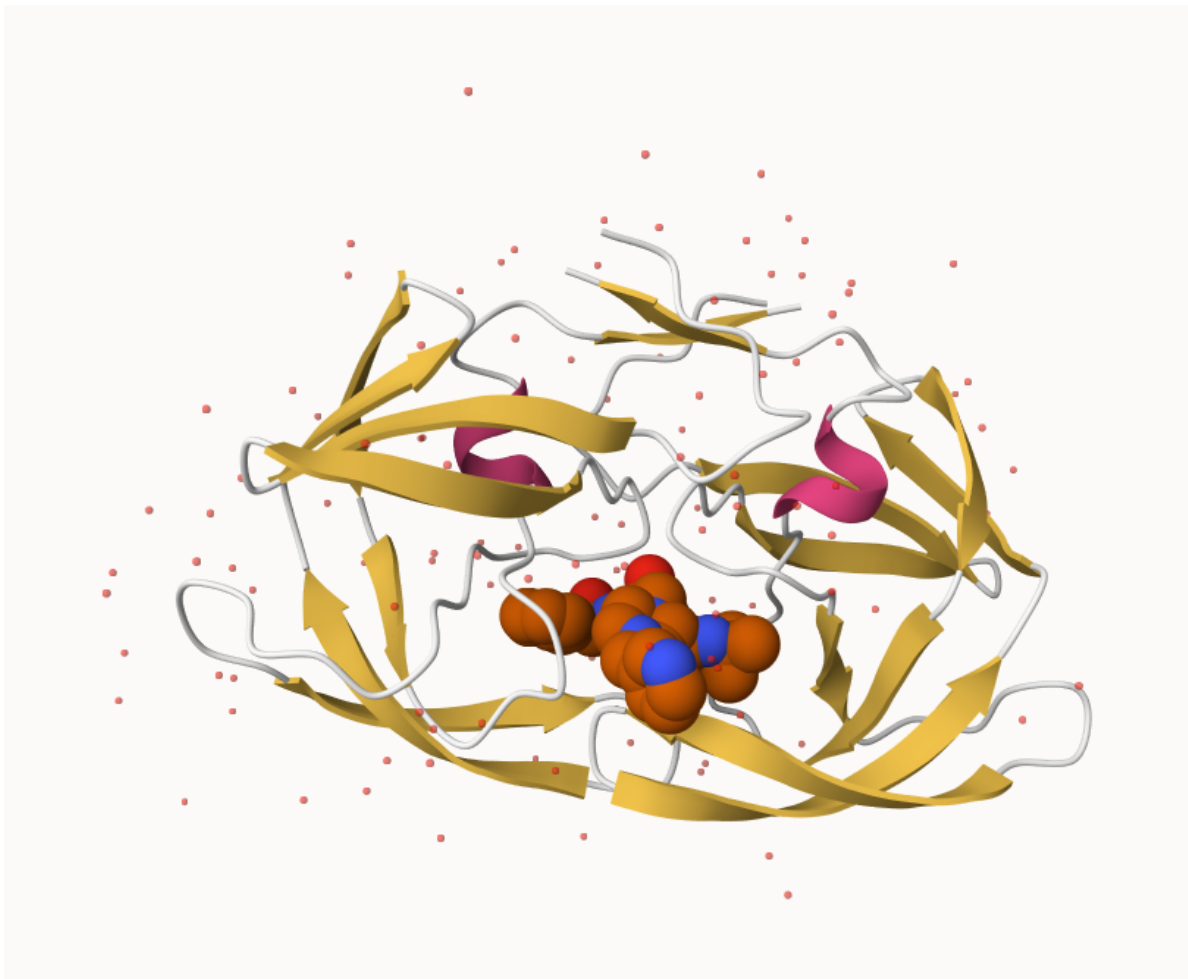
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

This has to do with how diffraction works. It likely can't identify the H molecule because of how small it is.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

It's hard to identify, but it looks like it is at around 3 angstroms. It looks like HOH 3##.

Q 6 Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



4 Introduction to Bio3D in R

```
library(bio3d)
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q 7 How many amino acid residues are there in this pdb object?

There are 198 AA residues. > Q8 Name one of the two non-protein residues?

They include water (HOH) and MK1. > Q9 How many protein chains are in this structure?

There are 2 (we also saw this in the image).

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

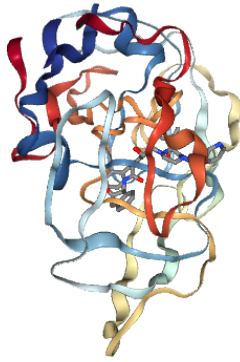
	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62

3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elemsy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

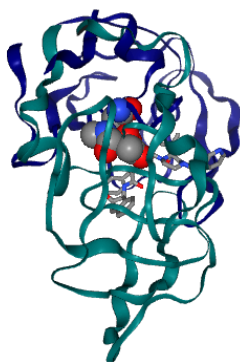
```
library(bio3dview)
library(NGLVieweR)

view.pdb(pdb) |>
  setSpin()
```



```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","teal"),
         highlight = sele,
         highlight.style = "spacefill") |>
setRock()
```

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

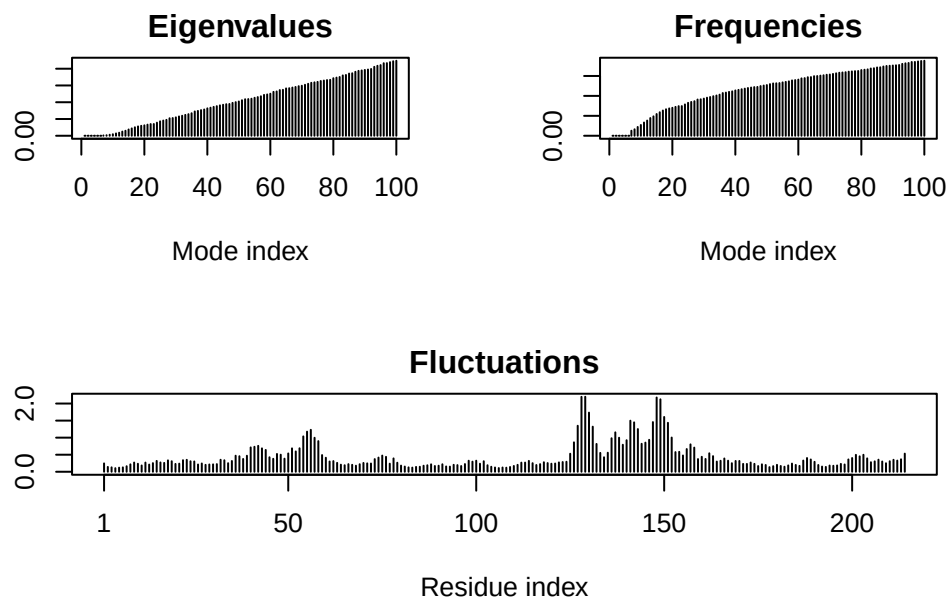
```
# Perform flexibility prediction
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.01 seconds.
```

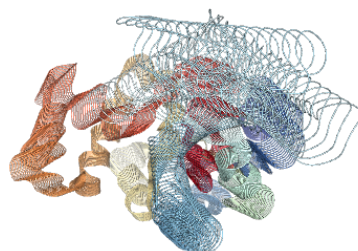
```
Diagonalizing Hessian... Done in 0.037 seconds.
```

```
plot(m)
```



```
mktrj(m, file="adk_m7.pdb")
```

```
view.nma(m, pdb=adk)
```



5 Comparative structure analysis of Adenylate Kinase

```
# Install packages in the R console NOT your Rmd/Quarto file
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa is only on BioConductor. > Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview is not on either

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

```
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60
      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120
      121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM TAPLIG
      121      .      .      .      .      .      180
      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDG TKPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

It appears the answer is 214.

```
hits <- NULL
```

```
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6H
```

```
# Download related PDB files
```

```
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/5EJE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):  
pdb/1E4Y.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/3X2S.pdb.gz exists. Skipping download

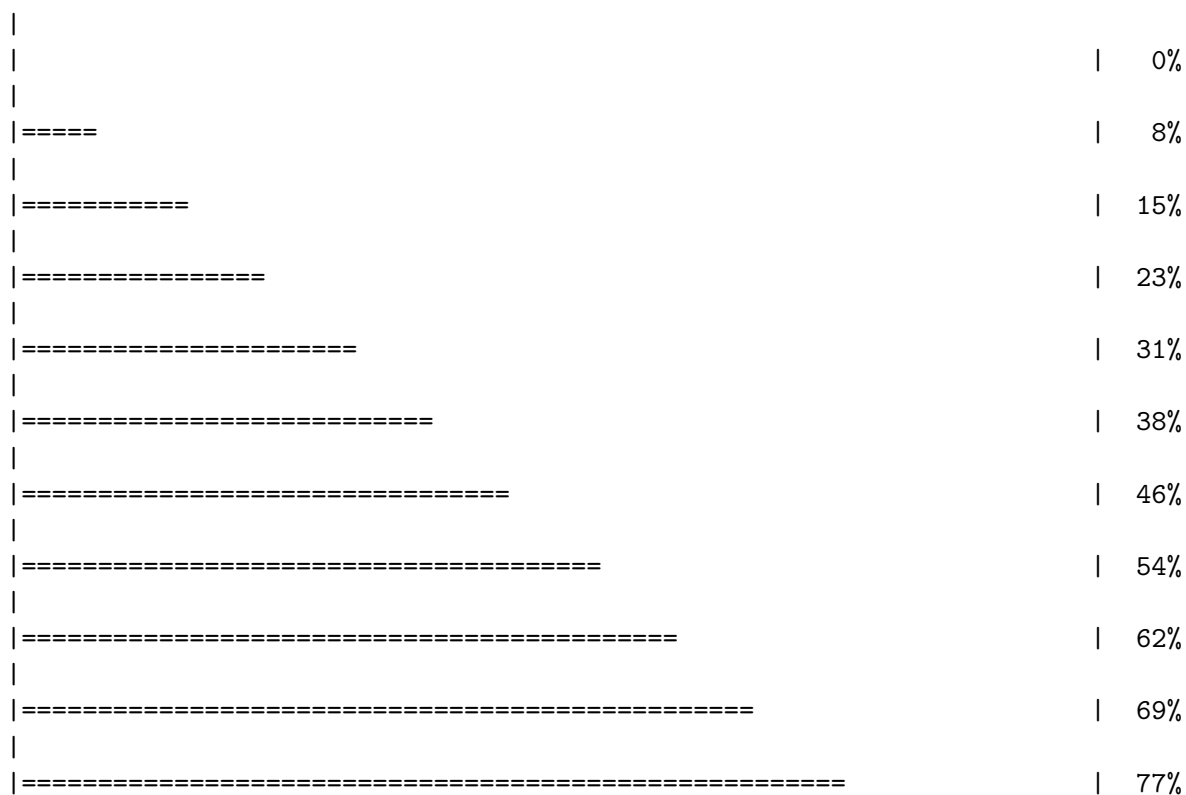
Warning in get.pdb(hits\$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6HAP.pdb.gz exists. Skipping download

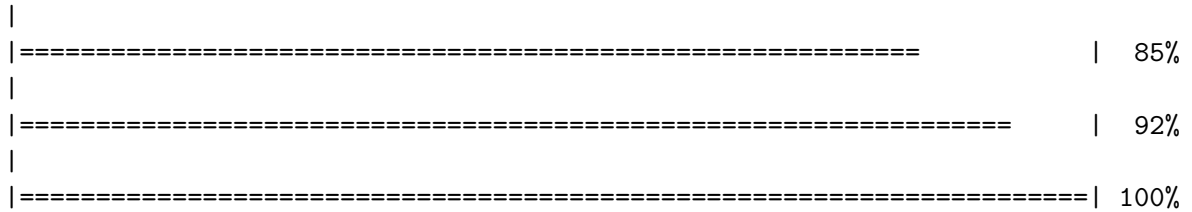
Warning in get.pdb(hits\$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4PZL.pdb.gz exists. Skipping download





Wednesday, February 11

Comparative protein structure analysis with PCA

We start with a database id “lake_A”

```
library(bio3d)

id <- "lake_A"

aa<- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa

pdb|1AKE|A      1      .      .      .      .      .      .      60
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
1      .      .      .      .      .      .      60

61      .      .      .      .      .      .      120
pdb|1AKE|A      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
61      .      .      .      .      .      .      120

121      .      .      .      .      .      .      180
pdb|1AKE|A      VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
121      .      .      .      .      .      .      180

181      .      .      .      214
pdb|1AKE|A      YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
181      .      .      .      214
```



```
Call:
  read.fasta(file = outfile)
```

```
Class:
  fasta
```

```
Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

```
blast <- blast.pdb(aa)
```

```
Searching ... please wait (updates every 5 seconds) RID = STP95NB4014
```

```
.
Reporting 96 hits
```

```
Have a wee peak:
```

```
head(blast$hit.tbl)
```

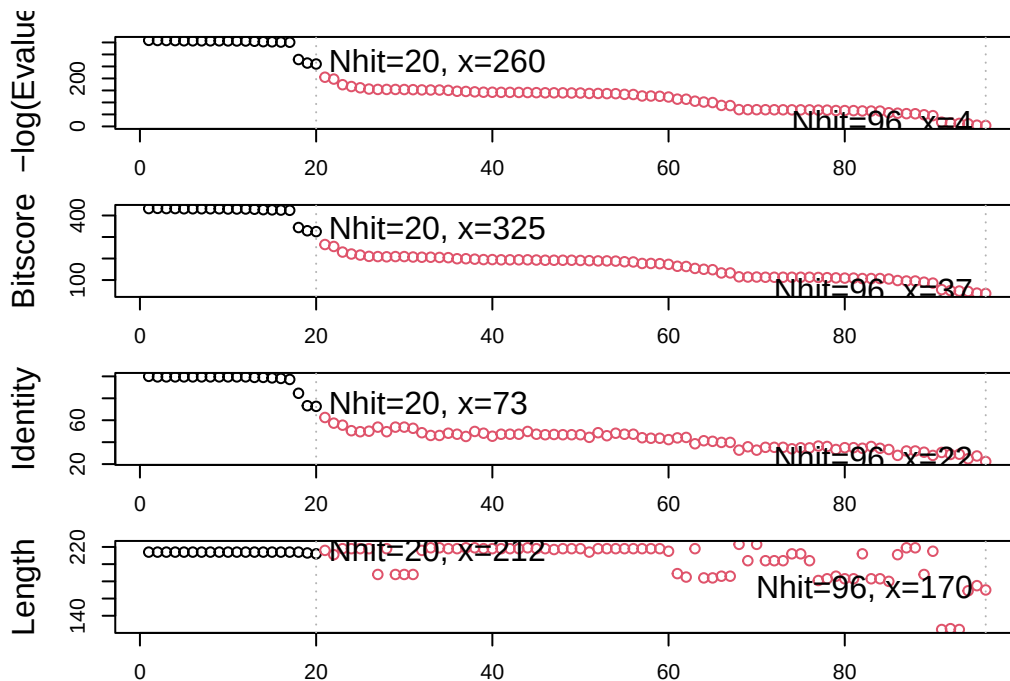
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_5314419	1AKE_A	100.000	214	0	0	1
2	Query_5314419	8BQF_A	99.533	214	1	0	1
3	Query_5314419	4X8M_A	99.533	214	1	0	1
4	Query_5314419	6S36_A	99.533	214	1	0	1
5	Query_5314419	9R6U_A	99.533	214	1	0	1
6	Query_5314419	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evaluate	bitscore	positives	mlog.evaluate	pdb.id	acc
1	214	1	214	1.78e-156	432	100.00	358.6267	1AKE_A	1AKE_A
2	214	21	234	2.91e-156	433	100.00	358.1351	8BQF_A	8BQF_A
3	214	1	214	3.18e-156	432	100.00	358.0464	4X8M_A	4X8M_A
4	214	1	214	4.67e-156	432	100.00	357.6621	6S36_A	6S36_A
5	214	1	214	1.04e-155	431	99.53	356.8615	9R6U_A	9R6U_A
6	214	1	214	1.23e-155	431	99.53	356.6937	9R71_A	9R71_A

```
hits <- plot(blast)
```

```
* Possible cutoff values:    260 3
    Yielding Nhits:         20 96
```

```
* Chosen cutoff value of:    260
    Yielding Nhits:         20
```



Peak ar our top hits

```
head(hits$pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

Now we can download these “top hits” these will all be ADK structures in the PDB database.

```
files <- get.pdb(hits$pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8M.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

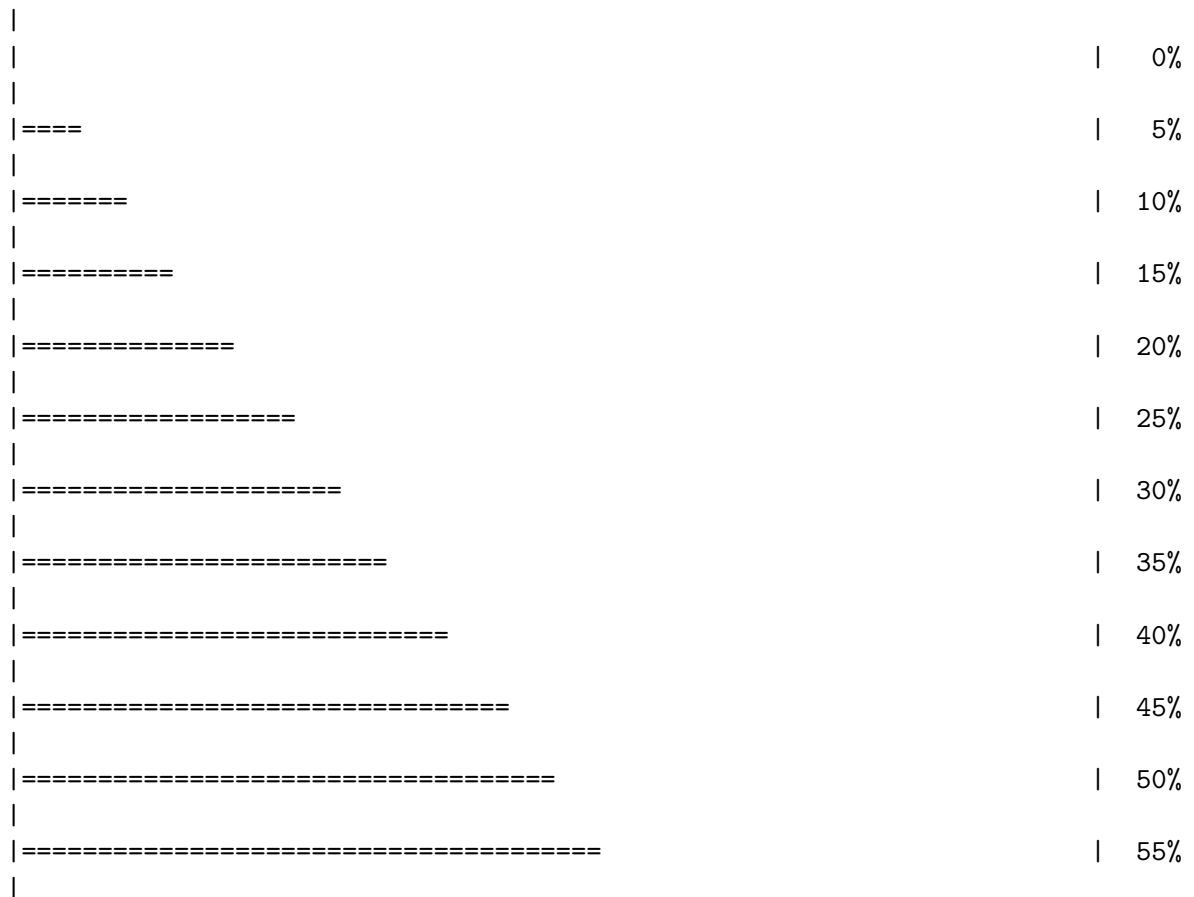
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

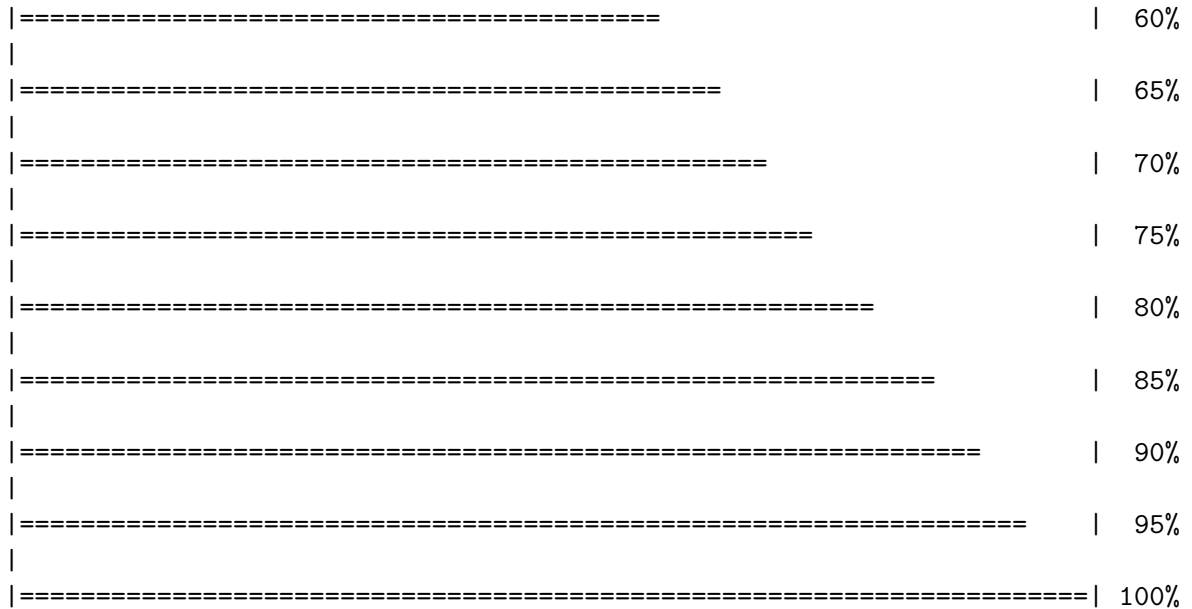
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download





We need one package from BioConductor. To set this up, we need to first install a package called “**BiocManager**” from CRAN.

Now we can use the `install()` function from this package like this: `BiocManager::install("msa")`

```
pdbbs <- pdbaln(files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
```

```

pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/8PVW_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/4NP6_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

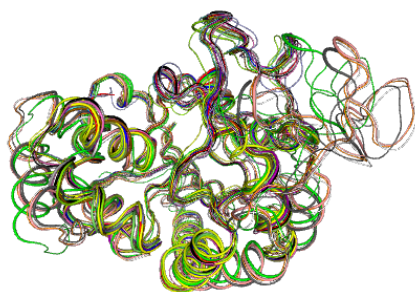
pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbs/split_chain/3HPR_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbs/split_chain/5EJE_A.pdb

```

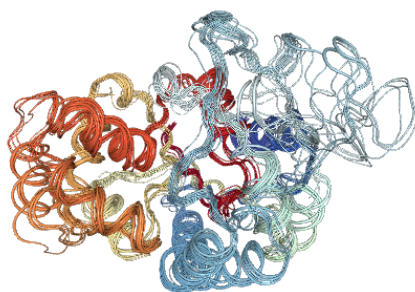
```
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20  name: pdbc/split_chain/4NP6_A.pdb
```

Let's have a wee peak at our structures after "fitting" or superposing:

```
library(bio3dview)
view.pdbc(pdbc)
```

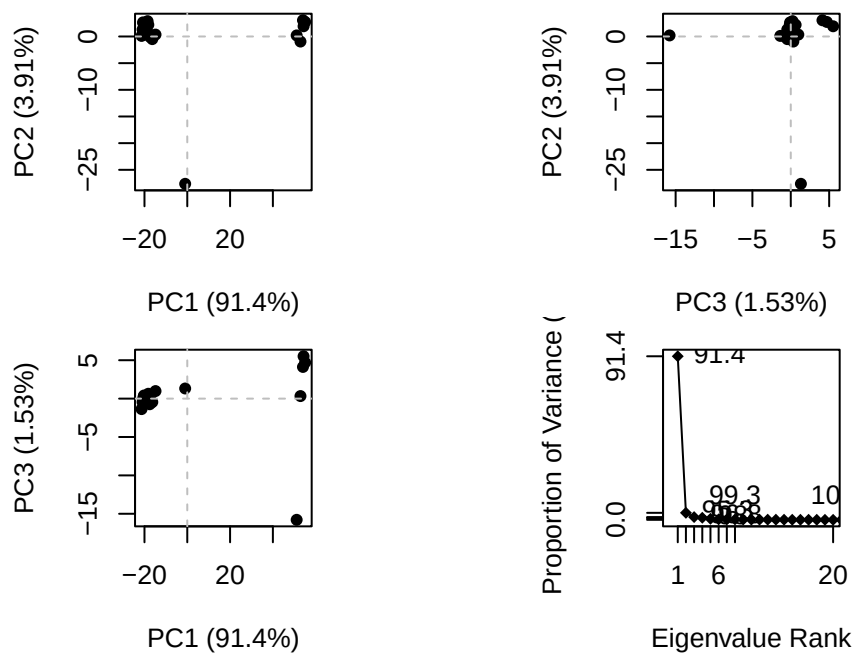



```
view.pdbs(pdb, colorScheme = "residue")
```

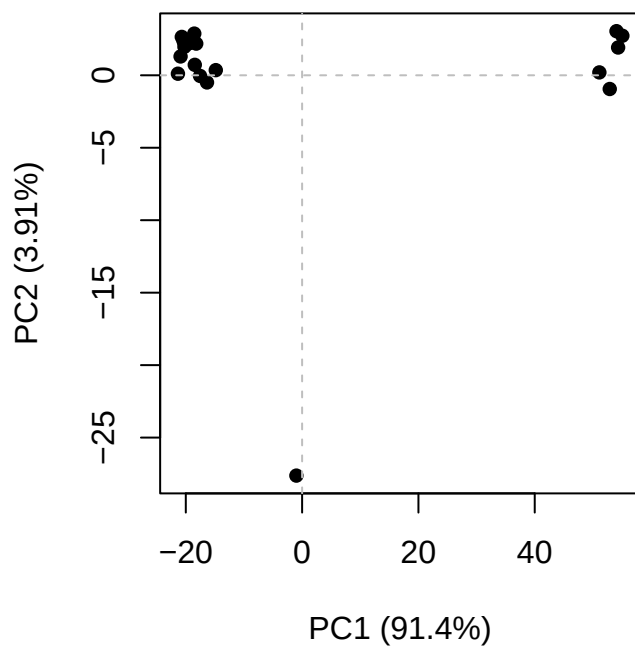


We can run functions like `rmsd()`, `rmsf()` and the best `pca()`

```
pc.xray <- pca(pdbbs)
plot(pc.xray)
```



```
plot(pc.xray, 1:2)
```



finally, let;s make a wee movie of the major “motion” or structural difference in the dataset - we call this a “trajectory”

```
mktrj(pc.xray, file="results.pdb")
```